

DEPARTMENT OF PHYSICAL SCIENCES IISER-MOHALI

IDC 402 - Nonlinear dynamics

Midsem-1

Due date: 05.02.2026, 3.00 PM

Date: 05.02.2026

Roll No.: [REDACTED]

Name: [REDACTED]

- Show all the steps of your calculations, marks will be deducted for missing steps.
- For questions involving graphs, label your axes properly.

Q 1. Consider the following dynamical system:

$$\dot{x} = x(1-x)(x-a),$$

(1)

where the parameter a satisfies: $0 < a < 1$

- Identify all fixed points (equilibria) of the system.
- Determine the stability of each fixed point using a phase flow analysis.
- Show that the dynamics can be interpreted as the motion of an overdamped particle in an effective potential, i.e.

$$\dot{x} = -\frac{dV}{dx}.$$

Derive the explicit form of the potential function $V(x)$ (up to an additive constant). Briefly explain how the stability of equilibria is reflected in the shape of the potential landscape $V(x)$. You may illustrate (by drawing the potential) your discussion using the example $a = 0.5$.

Q 2. Consider the following dynamical system

$$\dot{x} = x - \alpha \frac{x}{1+x^2}$$

where α is a parameter. Answer the following questions

- Determine the critical value α^* at which the bifurcation occurs, and construct a detailed bifurcation diagram. What type of bifurcation does the system exhibit?
- Around the bifurcation point, demonstrate that the system can be reduced to the normal form corresponding to the identified bifurcation.

Q 3. Consider the Romeo-Juliet system

$$\begin{aligned}\dot{R} &= aR + bJ, \\ \dot{J} &= bR + aJ,\end{aligned}$$

where $a < 0$ and $b > 0$.

- (a) Determine the fixed point of the system. 1
- (b) Classify all possible stability types of the fixed point, and sketch the corresponding phase portraits (approximate flow lines) for each case. 6

Useful Information

1. $\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots$, converges when $|x| < 1$.
2. How to plot/draw not so easy function $f(x)$? : Just choose some same points for x such as $x = -3, -2, -1, 0, 1, 2, 3$, find $f(x)$ and connect them to draw $f(x)$.